



Alvan Arulandu <aarulandu@college.harvard.edu>

Update on MOPS Algorithm

8 messages

Duy Thuc Nguyen <duythucn@mit.edu>

Sun, Jan 4, 2026 at 10:05 AM

To: Alan Edelman <edelman@mit.edu>

Cc: "aarulandu@college.harvard.edu" <aarulandu@college.harvard.edu>

Dear Prof. Edelman,

I hope you are doing well and wish you a happy new year. Over the past week, Alvan and I worked on the Maple code and reviewed the algorithm used to compute Jacobi polynomials. Alvan wrote a new Julia package to follow the Maple code closely and is easier to read and work with. More details about this Julia code will be included in our short report. I manually traced through the algorithm in Section 2.7.2 to test the univariate quadratic Jacobi polynomial case you sent earlier (screenshot attached), and the result matches the MOPS package. This suggests that there is a mismatch between the algorithm itself and Maple's default Jacobi polynomial. I have also written this up in the attached report.

Please let me know what you think would be the most appropriate next steps. I'm happy to continue studying/fixing the algorithm for the rest of winter break.

Best regards,
Duy Thuc

```
=
> simplify((a + b + 3) * (a + b + 4) * jacobi(2, [2], a, b, [x]))
      x^2 (4 + a + b) (3 + a + b) - 2 x (a + 2) (3 + a + b) + 2 a^2 + 6 a + 4
=
> simplify(2 * JacobiP(2, a, b, (1 - 2 x)))
      x^2 (4 + a + b) (3 + a + b) - 2 x (a + 2) (3 + a + b) + a^2 + 3 a + 2
=
```

MOPS.pdf
604K

Alan Edelman <edelman@mit.edu>

Sun, Jan 4, 2026 at 10:07 AM

To: Duy Thuc Nguyen <duythucn@mit.edu>

Cc: "aarulandu@college.harvard.edu" <aarulandu@college.harvard.edu>

hmm so the algorithm is itself broken?
this would require going back to the very start of it all ?
thank you for the hard work
[Quoted text hidden]

Alvan Arulandu <aarulandu@college.harvard.edu>

Sun, Jan 4, 2026 at 7:55 PM

To: Alan Edelman <edelman@mit.edu>

Cc: Duy Thuc Nguyen <duythucn@mit.edu>

Dear Prof. Edelman,

While I did find and fix errors in the implementation as documented in the pdf Duy Thuc sent previously, the algorithm is itself broken. However, it's unknown how far down this goes.

My first guess would be to look at the recursion for computing the Jacobi coefficients $c_{\{\kappa, \sigma\}}^{\alpha}$ in the paper (eq. 14, sec 2.7.2); it's still unclear to me how this recursion is derived.

Thank you for your advisement!

Sincerely,
Alvan
[Quoted text hidden]

Alan Edelman <edelman@mit.edu>

Mon, Jan 5, 2026 at 8:02 AM

To: Alvan Arulandu <aarulandu@college.harvard.edu>, Sungwoo Jeong <sjeong@cornell.edu>

Cc: Duy Thuc Nguyen <duythucn@mit.edu>

Sungwoo
Do u recall?
Id have to remind myself
[Quoted text hidden]

Sungwoo Jeong <sjeong@cornell.edu>

Mon, Jan 5, 2026 at 9:36 AM

To: Alan Edelman <edelman@mit.edu>, Alvan Arulandu <aarulandu@college.harvard.edu>

Cc: Duy Thuc Nguyen <duythucn@mit.edu>

Hi Alvan and Duy Thuc,

I would take a look at Ioana's thesis to find out where she came up with that Jacobi polynomial expression. In MOPS paper she does not give any reference, unfortunately.

Also, in the univariate case, would it also be possible to reduce the expression in 2.7.2 to a non-multivariate version?

It would be also interesting to compare that to known explicit forms of Jacobi polynomials (you can find some in Wikipedia, for instance), and find where it's incorrect.

Best,
Sungwoo

From: Alan Edelman <edelman@mit.edu>
Sent: Monday, January 5, 2026 8:02 AM
To: Alvan Arulandu <aarulandu@college.harvard.edu>; Sungwoo Jeong <sjeong@cornell.edu>
Cc: Duy Thuc Nguyen <duythucn@mit.edu>
Subject: Re: Update on MOPS Algorithm

[Quoted text hidden]

Alan Edelman <edelman@mit.edu>
 To: Sungwoo Jeong <sjeong@cornell.edu>
 Cc: Alvan Arulandu <aarulandu@college.harvard.edu>, Duy Thuc Nguyen <duythucn@mit.edu>

Mon, Jan 5, 2026 at 9:38 AM

<https://dspace.mit.edu/handle/1721.1/29347>

[Quoted text hidden]

Alan Edelman <edelman@mit.edu>
 To: Sungwoo Jeong <sjeong@cornell.edu>
 Cc: Alvan Arulandu <aarulandu@college.harvard.edu>, Duy Thuc Nguyen <duythucn@mit.edu>

Tue, Jan 6, 2026 at 9:07 AM

for the univariate case, stupid question (?), are you checking whether the interval is $[0, 1]$ vs $[-1, 1]$?
 as both are often used?

On Mon, Jan 5, 2026 at 9:37 AM Sungwoo Jeong <sjeong@cornell.edu> wrote:

[Quoted text hidden]

Duy Thuc Nguyen <duythucn@mit.edu>
 To: "aarulandu@college.harvard.edu" <aarulandu@college.harvard.edu>

Mon, Apr 20, 2026 at 11:25 AM

From: Alan Edelman <edelman@mit.edu>
Sent: Tuesday, February 3, 2026 1:31 PM
To: Duy Thuc Nguyen <duythucn@mit.edu>
Subject: Re: Update on MOPS Algorithm

yes still interested but got swamped

On Mon, Feb 2, 2026 at 11:10 PM Duy Thuc Nguyen <duythucn@mit.edu> wrote:

Hi Prof. Edelman,

I hope you are doing well. Since the semester has just begun and you are probably very busy with starting classes, and so am I, I was thinking perhaps it would be better if we find a time later this semester to discuss this project further, if this problem is still of interest to you, of course. Please let me know whenever is most convenient for you.

I will also most likely be around Harvard to do research this summer and would be happy to continue working on MOPS or perhaps the works on high-temperature limit of beta ensembles that I was part of my 18.338 project. Personally, being able to derive the connection with Bell polynomials for this new type of cumulants was very motivating for me and I would love to see if there are other connections. Thank you again for your help last Fall!

Best regards,
Duy Thuc

From: Duy Thuc Nguyen <duythucn@mit.edu>
Sent: Friday, January 30, 2026 11:38 AM
To: Alan Edelman <edelman@mit.edu>
Subject: Re: Update on MOPS Algorithm

Hi Prof. Edelman,

If this week doesn't work, I am also happy to find sometime this weekend or the next week. Please let me know when would be most convenient for you. Thank you in advance!

Best regards,
Duy Thuc Nguyen

From: Duy Thuc Nguyen <duythucn@mit.edu>
Sent: Sunday, January 25, 2026 8:49 AM
To: Alan Edelman <edelman@mit.edu>
Subject: Re: Update on MOPS Algorithm

Hi Prof. Edelman,

If Sunday doesn't work. I'm also happy to meet during the week. Please let me know what day(s) would work best for you. Thank you in advance for your help!

Best regards,
Duy Thuc Nguyen

From: Duy Thuc Nguyen <duythucn@mit.edu>
Sent: Thursday, January 22, 2026 5:52 AM
To: Alan Edelman <edelman@mit.edu>
Subject: Re: Update on MOPS Algorithm

Hi Prof. Edelman,

I should be free anytime before 4:30 PM this Sunday. Please let me know what works best for you. Look forward to the meeting and thank you a lot for your time!

Best regards,
Duy Thuc Nguyen

From: Alan Edelman <edelman@mit.edu>
Sent: Wednesday, January 21, 2026 9:38 AM
To: Duy Thuc Nguyen <duythucn@mit.edu>
Subject: Re: Update on MOPS Algorithm

Hmm maybe a zoom call getting into the weeds together

On Tue, Jan 20, 2026, 7:53 PM Duy Thuc Nguyen <duythucn@mit.edu> wrote:

Hi Prof. Edelman,

I hope you are doing well. Would you mind letting me know what is the next thing we should do about the MOPS algorithm and/or resources to look into?

Best,
Duy Thuc Nguyen

From: Duy Thuc Nguyen <duythucn@mit.edu>
Sent: Monday, January 12, 2026 10:56 PM
To: Alan Edelman <edelman@mit.edu>; sjeong@cornell.edu <sjeong@cornell.edu>
Subject: Re: Update on MOPS Algorithm

Hi Prof. Edelman and Prof. Jeong,

I hope you are doing well. I took a look at Dr. Dumitriu's thesis and the formulas that were important for the algorithm to compute Jacobi polynomials did not come with derivations either, similar to the paper. I was wondering if we just have to rederive these independently, or would there be other papers/resources that we can look into.

Best,
Duy Thuc

From: Duy Thuc Nguyen <duythucn@mit.edu>
Sent: Sunday, January 11, 2026 9:26 PM
To: Alan Edelman <edelman@mit.edu>
Subject: Re: Update on MOPS Algorithm

Hi Prof. Edelman,

Sorry for the late response. I fell sick the past couple of days and was unable to work. I just tried a range of possible values for the g_1 and g_2 parameters and they all failed the orthogonality test, as shown in the screenshot.

Best,

```

> with(MOPS)
arm, conjugate, expH, expHjacks, expI, expJjacks, expL, expLjacks, gbinomial, ghypergeom, gsfact, hermite, hermite2, issubpar, jack, jack2jack, jackidentity, jacobi, laguerre, leg, lhook, m2jack,
m2m, m2p, p2m, par, p, sfact, subpar, uhook]
> lab := proc(aa, bb)
local x, j3, j4;
j3 := jacobi(2/3, [2], aa, bb, [x]);
j4 := jacobi(2/3, [3], aa, bb, [x]);
evalf( Int( j3*j4*x*(1-x)^bb, x=0..1 ) );
end proc;
> aVals := [seq(i/4, i=0..4)]: # 0..1 by 0.25
bVals := [seq(j/4, j=0..8)]: # 0..2 by 0.25

M := Matrix(nops(aVals), nops(bVals),
(i, j) -> lab(aVals[i], bVals[j])) :

M;

```

$$\begin{bmatrix} -0.03333333333 & -0.01661305246 & -0.008840542174 & -0.004935529541 & -0.002857142857 & -0.001700405479 & -0.001033310124 & -0.0006373749985 & -0.0003968253968 \\ -0.03559070687 & -0.01755752535 & -0.009241487882 & -0.005099003093 & -0.002914269156 & -0.001710148573 & -0.001022954641 & -0.0006196705148 & -0.0003776507468 \\ -0.03626003626 & -0.01769310500 & -0.009203884727 & -0.005013727572 & -0.002825457371 & -0.001632064384 & -0.0009587379924 & -0.0005684986966 & -0.0003375206995 \\ -0.03572113605 & -0.01722620475 & -0.008847140112 & -0.004751996469 & -0.002636018670 & -0.001495330274 & -0.0008598542786 & -0.0004967018934 & -0.0002851425252 \\ -0.03428571429 & -0.01632389260 & -0.008266480994 & -0.004370571418 & -0.002380952381 & -0.001322100392 & -0.0007406168492 & -0.0004136754840 & -0.0002267573696 \end{bmatrix}$$

Duy Thuc

From: Alan Edelman <edelman@mit.edu>
Sent: Friday, January 9, 2026 10:41 AM
To: Duy Thuc Nguyen <duythucn@mit.edu>
Subject: Re: Update on MOPS Algorithm

does (0,0) work? is (1,0) giving (0,1)?
always like to find the smallest delta between what works and what doesn't

On Thu, Jan 8, 2026 at 10:34 PM Duy Thuc Nguyen <duythucn@mit.edu> wrote:

Yes, the checking of the univariate case was consistent with the Beta distribution on the interval [0, 1]. But just to be careful, I checked the integral on [-1, 1], and it still failed to satisfy the orthogonality relation.

```
> j3 := jacobi( 2/3, [2], 1, 2, [x] )
```

$$j3 := x^2 - \frac{6}{7}x + \frac{2}{7}$$

```
> j4 := jacobi( 2/3, [3], 1, 2, [x] )
```

$$j4 := -x^3 + \frac{4}{3}x^2 - \frac{3}{2}x + \frac{3}{7}$$

```
> int( j3*j4*x*(1-x)^2, x=0..1 );
```

$$-\frac{1}{4410}$$

```
> ff := j3*j4*x*(1-x)^2 :
evalf( Int(f, x=-1..1) );
```

$$-6.456235828$$

```
>
g1 := 1 : g2 := 2 :
t := 't':
```

```
j3t := expand(subs(x = (t + 1)/2, j3));
j4t := expand(subs(x = (t + 1)/2, j4));
```

```
# exact value
val_exact := int(j3t*j4t*(1+t)^g1*(1-t)^g2, t=-1..1);
```

```
# numerical value
val_num := evalf(val_exact);
```

$$\begin{aligned} j3t &:= \frac{1}{4}t^2 + \frac{1}{14}t + \frac{3}{28} \\ j4t &:= -\frac{1}{8}t^3 - \frac{1}{24}t^2 - \frac{11}{24}t - \frac{19}{168} \\ val_exact &:= -\frac{8}{2205} \\ val_num &:= -0.003628117914 \end{aligned}$$

Best,
Duy Thuc

From: Alan Edelman <edelman@mit.edu>
Sent: Thursday, January 8, 2026 10:05 AM
To: Duy Thuc Nguyen <duythucn@mit.edu>; Sungwoo Jeong <sjeong@cornell.edu>
Subject: Re: Update on MOPS Algorithm

so just to push , you were consistent with [0,1] when checking the univariate case?

i noticed wikipedia, for example, chooses [-1,1]

so when you say it doesn't get jacobi right in the univariate case,

i just wanted to be sure you use the same interval

On Thu, Jan 8, 2026 at 9:37AM Duy Thuc Nguyen <duythucn@mit.edu> wrote:

Hi Prof. Edelman,

Based on the screenshot sent by Sungwoo earlier, the orthogonality was tested on the interval $[0, 1]$.

Best,
Duy Thuc

From: Alan Edelman <edelman@mit.edu>

Sent: Tuesday, January 6, 2026 9:07 AM

To: Sungwoo Jeong <sjeong@cornell.edu>

Cc: aarulandu@college.harvard.edu <aarulandu@college.harvard.edu>; Duy Thuc Nguyen <duythucn@mit.edu>

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